

Interactive Knowledge Gardening: AI-Augmented Exploration in Immersive 3D Environments

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Abstract

We introduce Interactive Knowledge Gardening, an immersive knowledge-exploration system that blends the semantic strengths of large language models with 3D visualization in virtual reality, targeting the shortcomings of conventional search engines and purely text-based Artificial Intelligence (AI) interactions. Our approach visualizes information generated by an AI agent as a dynamically growing tree, which users actively cultivate using intuitive, natural interactions. In this metaphor, the primary query forms the tree's trunk, the key subtopics extend as branches, and detailed content grows as leaves at the periphery. Users can guide the AI agent to steer the exploration process, such as pruning" irrelevant branches or encouraging growth in areas of interest. Our proof-of-concept prototype shows that this embodied, spatial approach to information seeking can help users maintain contextual awareness while exploring complex topics.

CCS Concepts

· Human-centered computing~Human computer interaction (HCI)~Interaction paradigms~Mixed / augmented reality

Author Keywords

Immersive visualization; Knowledge investigation

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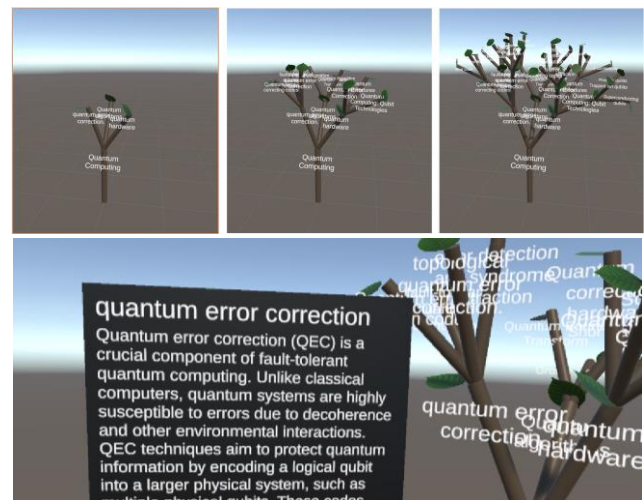


Figure 1: The Gardner Metaphor system visualizes a body of information collected by an AI agent into a dynamic virtual tree that grows over time (top row), while the user can interact with the system to guide and direct the investigation. The branches visualize a hierarchy of subtopics while the leaves represent content (e.g., text, image or video) relevant to that branch. Selecting a leaf will show the relevant content (bottom).

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1 Introduction

In today's digital era, being able to efficiently sift through the internet for meaningful information has become a crucial problem-solving skill. Traditional keyword-based search engines can overwhelm users with the sheer volume of unorganized results, often leading to cognitive overload. One approach to managing this complexity is to use visual knowledge-organization techniques like mind mapping, which

helps to structure and interlink ideas for better understanding. Mind maps, in particular, are a well-known tool that hierarchically organizes concepts and has been shown to improve learning and organizational skills [1]. Similarly, prior research in Augmented Reality (AR) explored how immersive technologies can provide more natural and intuitive ways to interact with information [2]. Early information visualization studies compared structured visual interfaces for search tasks, underlining the value of hierarchical representations in improving user comprehension [3].

Recent breakthroughs in large language models (e.g., ChatGPT [4]) now allow users to pose questions in everyday language and receive concise answers. However, these LLM-based services often lack intuitive visual feedback and sufficient user control, providing answers in text form with little sense of how those answers relate to a larger knowledge context.

To bridge this gap, we propose an interface that unites the rich semantic understanding of LLMs with immersive three-dimensional visualization to enable interactive, exploratory knowledge discovery. Our system, called the "Gardener Metaphor," lets users collaborate with an Artificial Intelligence (AI) agent to "grow" a visual knowledge structure in Virtual Reality (VR). As shown in Figure 1, the system represents knowledge hierarchies as dynamic tree structures where the main topic forms the trunk, subtopics grow as branches, and information items appear as leaves. The user can interact with the AI agent, either expanding the knowledge tree into new areas or refocusing it by pruning away less relevant growth. This system can be utilized for in-depth exploration by domain experts (e.g., researchers, students) who require data exploration and brainstorming.

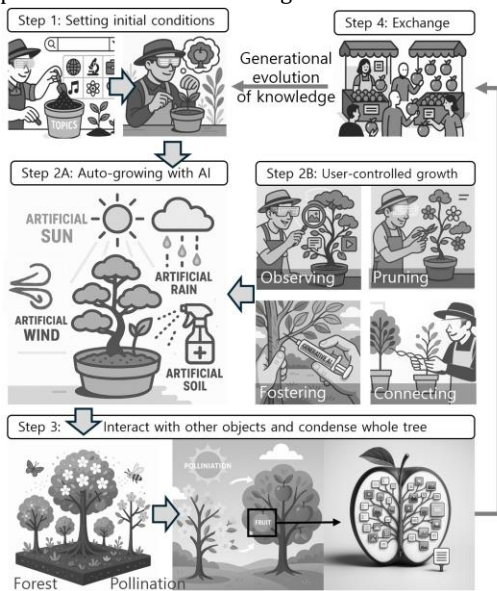


Figure 2: A conceptual illustration of the Gardener Metaphor (subfigures partially generated by ChatGPT).

Our approach builds on prior research in AI-assisted visualization and immersive learning. For example, it aligns with Wang et al.'s VIS+AI framework for augmenting data exploration with AI [5], and with insights from Fernández et al.'s survey of eXtended Reality in education [6]. It also builds upon KnowledgeNavigator by Yoon et al. [7], which structures query results into hierarchies of subtopics. However, while such systems present information on 2D displays, our work embeds the AI agent and knowledge structure in a fully immersive 3D space where information can be directly manipulated. Recent research has introduced LLM-assisted visual interfaces for knowledge graph exploration[10], experimentation platforms for large language models[11], and the integration of LLM-based AI agents within immersive virtual reality environments[12], collectively advancing interactive and intelligent information systems. This bridges the gap between intelligent information retrieval and natural user experience, creating a more engaging and cognitively supportive way to explore knowledge.

2 System Design

Our Interactive Knowledge Gardening (hereinafter referred to as the gardening metaphor) system consists of an immersive visualization environment and an AI-driven backend that work together to generate and refine the knowledge tree. In this section, we detail the design of the knowledge tree metaphor and the implementation of user interactions and AI processes that realize the gardening paradigm as summarized in Figure 2.

2.1 Knowledge Tree Visualization

Upon an initial user query, the system initiates a new knowledge tree in the virtual environment. The trunk represents the primary topic (the user's question), with first-level branches denoting major subtopics or facets, generated by the LLM agent that analyzes the query and identifies key areas for exploration. Branches can spawn sub-branches as AI-collected information expands into finer subtopics, forming a hierarchical structure. Leaves at branch ends contain detailed content, such as summary paragraphs, facts, images, or video clips related to the subtopic.

The tree is visualized in a three-dimensional (3D) environment accessed via VR headset. The tree metaphor was selected for its structural clarity and metaphorical depth, organizing information hierarchically: general concepts near the trunk, specific details at the leaves. This extends concept and mind maps into immersive 3D. Users navigate spatially by looking around or approaching branches to examine leaves. Visual cues like color and size add meaning—for example, fostered branches appear greener and thicker, while pruned ones wither. Keyword labels on branches and leaves offer quick context, and selecting a leaf displays full content on a virtual panel.

The tree grows dynamically, incorporating new branches and leaves as the AI uncovers relevant information or responds to follow-up queries. A new query plants a separate tree, allowing users to build a garden of multiple trees representing related

topics. Spatial arrangement enables exploration of inter-topic relationships, such as placing similar trees closer or highlighting connections like vines. This evolves beyond single-query results into a persistent, explorable knowledge base, leveraging 3D space for large-scale mapping unattainable on flat screens.

2.2 Human-AI Interaction and Collaboration

A key aspect of our system is the continuous collaboration between the human user and the AI agent. The AI agent is responsible for collecting information and generating content as it is visualized as the growth of the tree over time. Meanwhile, the user stays in control of directing this growth via their inputs – for example, using voice commands or hand gestures (tracked by the VR system) to influence what the AI focuses on.

Using speech, the user can issue commands or ask follow-up questions to the AI agent. For example, the user might point at a particular branch and say, “I want to know more about this,” which the system interprets as a request to expand that subtopic, and triggers the LLM to delve deeper into that branch’s subject, generating additional sub-branches or leaves with finer details. Conversely, the user might say “This branch is not relevant,” and the system will prune that branch away – effectively telling the AI to steer the exploration in a different direction. This spoken dialogue with the system helps maintain a natural flow of exploration, as if conversing with a knowledgeable assistant, while the virtual tree keeps track of context.

Compared to speech input, hand gestures can provide a more direct manipulation of the knowledge tree. The user can reach out to grab a branch and prune it, removing that branch and its leaves from the tree. The user may also indicate interest in growing a particular branch more using a “fostering” gesture (e.g., holding their palm up under a branch or making a pouring motion). We introduce a set of metaphoric gestures that map to commands: pruning (cutting motion), fostering (pouring motion), connecting branches (bringing two branches together to suggest a link), and so on. When these gestures are made, the corresponding command is executed by the system. For instance, upon a prune gesture, the system not only visually removes the branch but also informs the AI agent to cease further exploring that subtopic in subsequent information gathering. In contrast, a fostering gesture on a branch causes the AI agent to prioritize expanding that part of the tree further – analogous to giving extra resources to that “plant”.

Behind the scenes, the AI agent continually performs a cycle of information retrieval and content generation. We utilize an LLM which can search for information and generate summaries. When the user first asks a question, the LLM identifies initial subtopics by retrieving relevant sources (web pages, articles, etc.) and summarizing key aspects into branch content. For each branch or follow-up query, the LLM fetches more detailed information. This iterative content-generation process allows the knowledge tree to progressively grow, adding layers of detail over time. Importantly, if left unguided, the AI will autonomously explore the query topic in breadth and depth to suggest new branches the user might not have explicitly asked for. In this sense, the system can surface

unexpected connections or subtopics – akin to a gardener noticing a new sprout – which the user can then either cultivate or remove. Thus, in our system the human and AI engage in a balanced partnership: the AI contributes knowledge breadth and depth, and the human contributes judgment, intuition, and direction.

To keep the user informed of the AI agent’s activities, subtle cues can be provided. For example, when the AI is researching and growing the tree, new branches would appear with a gentle glow or a growing animation, so the user can immediately see where expansion is happening. Audio feedback or a brief narration can also accompany introduction of new leaves (e.g., “adding details about X...”). If the AI agent encounters contradictory information or uncertainty, it might create a special marker (like a leaf with a warning color) to invite the user to review it, thus maintaining transparency. Our goal is for the user to feel that they are working together with the AI agent to cultivate the knowledge garden, rather than just passively observing an AI-generated visualization.

3 Prototype Implementation

As a proof of concept, we developed a prototype system that implements the Gardener Metaphor on readily available hardware and software platforms. Figure 3 provides an overview of the system architecture and interaction pipeline. The prototype was built using the Unity 2021.3 game engine to create the VR environment, with deployment targeted to the Meta Quest 3 headset for standalone wireless operation. We leveraged the Oculus Integration toolkit (including hand-tracking support) to capture user gestures and the Unity XR Interaction Toolkit to manage basic VR interactions (like pointing and UI gaze). Voice commands are captured via the headset’s microphone and processed using a speech-to-text service, after which they are sent to the LLM module.

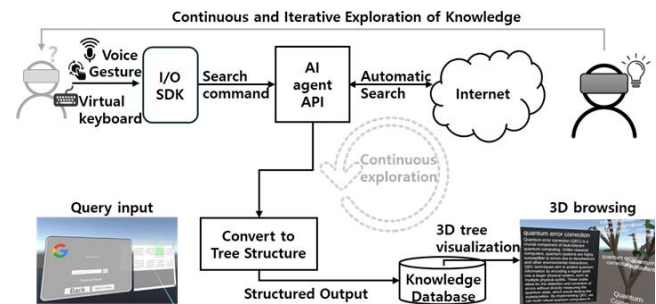


Figure 3: System architecture and interaction pipeline of Interactive Knowledge Gardening

The LLM processing is handled by calls to the Google Gemini API. In our implementation, when the user’s query is received, the system prompts the LLM with a custom instruction to act as a “knowledge gardener” – i.e., to output an organized breakdown of the topic into subtopics and generation of their relevant content. The Unity application progressively request the LLM to provide and expand the branches of subtopics and their relevant content.

The response from LLM is parsed and used to visualize the corresponding branches and leaves onto the virtual tree. Each branch is labeled with the subtopic title, and each leaf is associated with content that can be viewed when selected.

As the user interacts with the system, subsequent calls to the LLM are made. For example, if the user fosters a particular branch, subsequent query about the relevant topic is requested to the LLM to add further details. On the other hand, if the user prunes a branch or indicates it is irrelevant, the system will include a negative bias in future prompts (e.g., “Exclude information about [pruned topic].”). In this manner, the user’s real-time feedback dynamically shapes the LLM’s ongoing content generation.

Managing the performance and latency of this pipeline is crucial for a smooth user experience. We implemented the tree growth to be incremental and non-blocking. When a new query is sent to the LLM, a placeholder “growing branch” is shown on the tree (like a budding branch) so the user knows the system is working, and once the data returns (which typically takes a few seconds), they are added. This approach prevents the user from feeling stuck waiting; they can continue examining other parts of the tree in the meantime. We also set limits on the tree’s expansion at any given time to avoid overwhelming the user with too much information at once.

4 Discussion

Even in this early prototype, navigating and manipulating the knowledge tree showed promising potential. Users could explore ideas from multiple perspectives, gaining broader context when stepping back, or examining specific details when moving closer. Interacting with information through hand interaction and motion (such as removing branches of misinformation) appeared to offer a relatively intuitive and engaging experience.

The development of the Gardener Metaphor system revealed significant strengths and challenges. Its primary strength lies in transforming abstract search processes into tangible experiences. Test users reported that visualizing information as a growing tree helped maintain exploration context, while immersive gardening actions made learning feel creative and enjoyable. Additionally, the human-AI collaborative model enabled users to either accept AI agent’s suggestions or redirect them, fostering trust and agency.

However, several challenges emerged. Managing information overload in VR environments is difficult, as overly complex tree structures can become navigation barriers. We partially address this through growth rate controls and pruning capabilities. The accuracy of AI-generated content remains concerning, as visualization might reinforce LLM errors, highlighting the need for source transparency and verification mechanisms. VR-specific usability issues include gesture learning curves and headset fatigue, suggesting the need for clear tutorials.

Despite these challenges, the Gardener Metaphor presents a compelling vision for future information interfaces. It shifts from simple query-response interactions to exploratory partnerships between users and AI. By exposing the landscape of possible insights rather than single answers, the system makes AI more

transparent and controllable while externalizing users’ cognitive processes through visual representations of their inquiry paths.

5 Conclusion and Future Work

This paper introduced Interactive Knowledge Gardening, a novel system that enables users to explore information by collaboratively growing virtual knowledge trees in VR with AI assistance. By combining hierarchical knowledge structuring with intuitive embodied actions, users maintain agency while benefiting from AI’s information gathering capabilities.

Future development will focus on five key areas: (1) enabling multi-user collaborative exploration through both synchronous and asynchronous knowledge sharing; (2) implementing adaptive cognitive support that responds to users’ expertise levels and cognitive states; (3) developing cross-platform continuity to access knowledge trees across VR, PC, and mobile devices, including gesture recognition and multi-modal interface features; (4) creating an emotionally intelligent AI assistant that responds to users’ affective signals; and (5) introducing automated summarization and pruning to maintain clear visualizations as trees grow.

We plan user studies involving complex search and learning tasks using the Interactive Knowledge Gardening system. We will observe how users adapt to the gardening metaphor and whether it helps maintain context and deepen understanding. Following recommendations from prior work on evaluating immersive learning tools[2], the study will measure how effectively the spatial metaphor aids memory (e.g., recall tests) and how interactivity affects users’ sense of control and engagement. This study will provide insights to guide interface refinements, ensuring the design benefits users. Formal evaluations will compare this system with traditional interfaces on recall, comprehension, and engagement.

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